Ruhaan Bhargav

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Education

Purdue University West Lafayette, IN
B.S. in Computer Science and B.S. in Mathematics, GPA: 3.74/4.0 *Aug. 2024 – May 2028*

- **Relevant Courses:** Data Structures & Algorithms, Systems Programming, Linear Algebra, Multivariate Calculus, Real Analysis, Differential Equations, Statistics

Experience

BoilerQuant Jan. 2026 – Present
Quantitative Analyst *West Lafayette, IN*

- Member of a collaborative student organization focused on studying and applying quantitative finance concepts including statistical modeling, time-series analysis, and algorithmic strategy development
- Co-authoring research about systematic favorite–longshot bias in mid-size prediction markets (e.g. Kalshi) and developing frameworks to convert identified mispricings into actionable equity strategies

Purdue University – CS Department Aug. 2025, Mar. 2026 – Present
Student TA → Developer TA (rehired) *West Lafayette, IN*

- Served as a TA for Purdue's intensive 2-week CS Bridge crash course (Aug. 2025), supporting students in foundational programming and problem-solving
- Rehired as Developer TA: authoring challenge problems, developing course material, and building automated test suites using JUnit to support the CS Bridge curriculum

Purdue Orbital Jun. 2024 – Dec. 2025
Avionics Engineer *West Lafayette, IN*

- Designed and implemented embedded sensor packages (accelerometers, barometer, IMU) for rocket and high-altitude balloon telemetry from scratch using Rust and Embassy; published code adopted externally
- Established a real-time inter-module data pipeline across GNC, Propulsion, and Systems teams using CAN bus protocol, gaining hands-on experience with low-latency, high-reliability data transport

Projects

High-Performance Order Book Matching Engine | *Python, Alpaca API, NumPy* Nov. 2025 – Present

- Building a low-latency order book simulator processing 10,000+ orders/second with price-time priority matching, benchmarking multiple priority queue data structures for optimal throughput
- Implementing multivariate Hawkes processes to model self-exciting order flow, capturing realistic order clustering and volatility dynamics observed in live equity markets
- Integrating Alpaca trading API for real-time SPY order book data to validate simulated outputs against live market conditions; leveraging results to design and evaluate optimal execution strategies

Lead-Lag Pairs Trading Strategy | *Python, yfinance, pandas, scipy, seaborn* Feb. 2026

- Investigated time-lagged correlations between large-cap tech leaders (AAPL, MSFT, NVDA) and mid-cap laggers (AMD, INTC, MU) using 3 years of daily price data fetched via yfinance
- Built a statistical pipeline computing Pearson correlations with p-value significance testing across all 9 leader-lagger pairs and lag windows of 1–7 days to identify exploitable lead-lag relationships
- Designed a trading strategy that enters positions in lagging stocks upon leader movement, analyzing optimal lag periods and correlation decay to maximize risk-adjusted returns

Custom Ticketing Platform – Purdue Grand Prix | *Python, SQLite, QR, Cryptography* Dec. 2025 – Present

- Built a full-stack ticketing system with a SQLite backend, QR code generation for contactless entry, and a secure authentication layer using cryptographic hashing and custom ID generation
- Designing a modular architecture to allow extensions including API integration for database scaling, and payments

Technical Skills

Languages: Python, C/C++, Java, Rust, JavaScript, C#, R, HTML/CSS, SQL
Libraries & Frameworks: PyTorch, pandas, NumPy, matplotlib, seaborn, scipy, Next.js, React.js, JUnit, Embassy
Tools & APIs: Git, SQLite, LaTeX, Linux, Alpaca API, yfinance, REST APIs
Spoken Languages: English (Primary), Mandarin (Conversational), Hindi (Native), Punjabi (Native)